

Homework 2: Policy Gradient vs Value-Based RL in JAX

Due: March 17, 2026

Submission: Brightspace Homework 2

Objective

Understand algorithmic differences between value-based and policy-based reinforcement learning methods, and leverage JAX for efficient implementation. Explore how environment design influences research questions and results.

Tasks

1. Create a Grid-World Environment

- Provide a simple environment where an agent navigates a grid to achieve a goal.
- **General Guidance:**
 - Define states as grid positions and actions as movements (e.g., up, down, left, right).
 - Include rewards for reaching the goal and penalties for undesirable states (e.g., obstacles).
 - **Encouragement:** Put your own spin on the environment! Consider how its design relates to the research questions you want to explore (e.g., sparse rewards, stochastic transitions, obstacles).
- Document your design choices and rationale in the report.
- **Gymnasium Compatibility Note:**
 - Use the **Gymnasium** framework for your environment implementation.

- Ensure that:
 - Observations and rewards are converted to JAX arrays using `jnp.asarray()` before processing.
 - For batching or vectorized rollouts, you may need custom wrappers since Gym environments are stateful and not inherently JAX-native.
- **Optional Advanced Exploration:**
 - Consider the use of **Gymnax** (JAX-native Gym environments) or **Brax** (differentiable physics engine for RL) in-place of Gymnasium.
 - These libraries provide:
 - Fully functional, immutable environments aligned with JAX's paradigm.
 - Built-in support for vectorized rollouts and GPU/TPU acceleration.
 - Opportunities to experiment with differentiable environments and large-scale RL.

2. Implement and Train Agents

- **Random Agent (baseline)**
- **DQN (value-based)**
- **REINFORCE (policy gradient)**
- Implemented in **pure JAX** using `jit` for speed.
- Use **vmap** for batched rollouts where possible.

3. Experimentation

- **Hyperparameter Search Guidance:**
 - Explore at least **3 learning rates** (e.g., 0.001, 0.01, 0.1).
 - Explore at least **2 discount factors** (e.g., $\gamma = 0.9, 0.99$).
 - Document which combination performs best and why.
- **Multiple Independent Seeds:**
 - Run **10 independent seeds** for each algorithm and hyperparameter setting.
 - Seeds should affect initialization and environment randomness.
- **Plotting Requirements:**

- **Plot 1:** Mean and standard error of reward vs episodes across 10 seeds for each algorithm (use shaded error bands or error bars).
- **Plot 2:** All 10 individual seed runs overlaid for each algorithm (to show variability).
- Compare learning curves for Q-learning, REINFORCE, and baseline agent.
- Visualize final policies using **heatmaps** and/or **action arrows** for your grid-world.

Deliverables

- **Code:**
 - JAX implementations of Q-learning and REINFORCE.
 - Grid-world environment code.
- **Report (PDF):**
 - Description of your environment design and rationale.
 - Learning curves for all agents (including mean $\pm$ standard error and individual runs).
 - Policy visualization (heatmaps, arrows).
 - Hyperparameter sensitivity analysis.
 - Reflection on:
 - The mean performance of your agent versus the variability across seeds. If there is a high-degree of variability discuss the potential causes and how you could assess this.
 - How JAX's functional style influenced your implementation.
 - Relationship between environment design and research questions.
 - GPU usage details (Morningstar, Colab, or other).
 - AI tool usage (if applicable): *What did AI suggest? What did you accept/reject and why?*

GPU Requirement

- You must run experiments on a GPU:
 - **Morningstar cluster** (university resource)
 - Personal GPU
 - **Google Colab** or other cloud-based GPU services
- Document your setup in the report.

Grading Rubric (100 points)

Component	Points
Environment Design & Rationale	20
- Creativity and alignment with research goals	10
- Clear documentation	10
JAX Implementations	20
- Correct Q-learning implementation	10
- Correct REINFORCE implementation	10
Analysis & Visualization	30
- Learning curves (mean $\pm$ SE + individual runs)	10
- Policy heatmaps/arrows	10
- Hyperparameter sensitivity	10
Report Quality	30
- Description of motivation and alignment of environment / task	10
- Description of experimental set	10
- Analysis of results and reflection	10

Implementation Guidance

Basic implementation guidance is provided below. For more detailed guidance see:

Deep Reinforcement Learning Hands-On A practical and easy-to-follow guide to RL from Q-learning and DQNs to PPO and RLHF

Lapan, Maxim

2024

https://ocul-uo.primo.exlibrisgroup.com/permalink/01OCUL_UO/5lqjs2/alma991048019603105161